CONCESSIONS PRICING CHALLENGE · CAVS SPORTS BUSINESS & ANALYTICS NIGHT 2026

SHOW PLANNER

Place any show in one of five crowd groups, forecast what its fans will buy, and test prices before the doors open.
Try it live: zeynepbastas-cavs-show-planner.streamlit.app

Blessing Masanga **Daniel Youngquist** **Zeynep Bastas**

Case Western Reserve University

🟩 Family Day 🟧 Family Night & Sports 🟪 Pop 🟨 Social 🟦 Bar

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Place any show in one of five crowd groups, forecast what its fans will buy, and test prices before the doors open. Every number updates as you type, starting from the choices our team made.

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5 crowd groups · Crowd: Rule 2 base case · Prices: our card · Elasticities: published studies

THE SHORT VERSION

1. **Five crowd groups, one simple rule.** A sports event, or a family show where most buyers are men: Family Night & Sports. Other family shows: Family Day. Rock or country concert: Bar. Pop concert: Pop. Everything else: Social.
2. **The groups make forecasts more accurate.** Predicting each act from the other shows in its group cuts the error from 34% (no groups) to 22%.
3. **Each crowd buys differently.** Bar crowds spend 7 times more on alcohol per fan than Family Day ($13.30 vs $1.80). Pop fans buy the most water, while hip-hop and comedy crowds lean on cocktails.
4. **Prices flex by group.** Water $4.99 everywhere and family prices for both family groups. Adult crowds pay a little more for drinks: cocktails $30.25 and Michelob $17.49 at Pop and Social shows, and $30.99 and $17.99 at Bar shows.
5. **For the 9 upcoming shows:** $1,055 more than at the old prices (+0.5%) and 4.0% more items sold, at our card and the base crowd. The card stays ahead of the old prices at every price sensitivity we tested, from 0x to 2x.

MEET THE FIVE CROWDS

36 past shows; "per fan" means per ticket sold

● Family Day 7 past shows

Family shows where fewer than half the buyers are men

$1.80 on alcohol and **$7.70** on all concessions per fan

Per 1,000 fans: 95 waters, 10 cocktails, 71 fountain drinks, 61 pretzels, 10 beers

Our prices: Water $4.99 · Cocktail $22.99 · Fountain $6.49 · Pretzel $7.49 · Michelob $12.99

● Family Night & Sports 5 past shows

Sports events, and family shows where at least half the buyers are men

$6.10 on alcohol and **$12.90** on all concessions per fan

Per 1,000 fans: 118 waters, 30 cocktails, 81 fountain drinks, 61 pretzels, 42 beers

Our prices: Water $4.99 · Cocktail $22.99 · Fountain $6.49 · Pretzel $7.49 · Michelob $12.99

● Pop 5 past shows

Pop concerts

$8.50 on alcohol and **$13.10** on all concessions per fan

Per 1,000 fans: 168 waters, 48 cocktails, 58 fountain drinks, 46 pretzels, 27 beers

Our prices: Water $4.99 · Cocktail $30.25 · Fountain $7.79 · Pretzel $10.59 · Michelob $17.49

● Social 10 past shows

Hip-hop/R&B, comedy and other adult shows

$9.50 on alcohol and **$12.90** on all concessions per fan

Per 1,000 fans: 114 waters, 71 cocktails, 37 fountain drinks, 30 pretzels, 29 beers

Our prices: Water $4.99 · Cocktail $30.25 · Fountain $7.79 · Pretzel $10.59 · Michelob $17.49

● Bar 9 past shows

Rock and country concerts

$13.30 on alcohol and **$17.60** on all concessions per fan

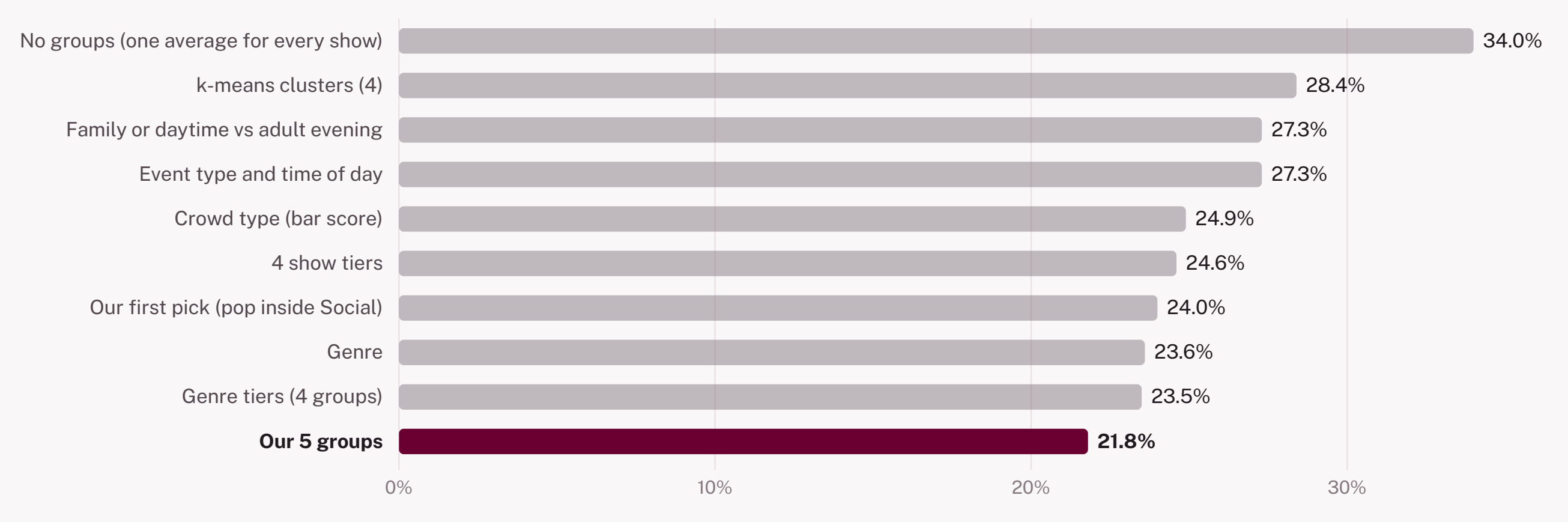
Per 1,000 fans: 163 waters, 82 cocktails, 43 fountain drinks, 42 pretzels, 80 beers

Our prices: Water $4.99 · Cocktail $30.99 · Fountain $7.79 · Pretzel $10.59 · Michelob $17.99

HOW WE PICKED THE GROUPS

Backtest error, lower is better

For each way of grouping the 36 past shows, we hid every show by one act, predicted its sales of the five items from the other shows in its group, and compared with what it really sold. Our five groups had the lowest error of these ten. We also tried a few add-ons (group totals instead of averages, demographic adjustments, a different grouping for each item). The best were only about 1 point better (20.5% at best), and with 36 shows a gap that small can be luck, so we kept the simple rule.



GROSS, 9 SHOWS, 5 ITEMS
$221,923
+0.5% vs the old prices ($220,868)

ITEMS SOLD
20,663
+4.0% vs the old prices

CROWD, ALL 9 SHOWS
62,740
Base case (Rule 2)

GROSS RANGE, LOW TO HIGH CROWD
$195K to $276K
At your prices

CROWD SIZE
Low | **Base** | High

PRICE SENSITIVITY
Half | **As published** | 1.5x

DOWNLOAD THE CSVS
Save upcoming_forecast.csv | Save event_groups.csv
Saves what is on screen. Our submitted files use Our card, Base and As published.

PRICE CARD

Start from: Our card | What each group paid | Current menu everywhere

Type a price in any cell. The small number is the change against what that group paid on average in past shows.

ITEM	● Family Day	● Family Night & Sports	● Pop	● Social	● Bar	CURRENT MENU
20oz **Dasani Water**	4.99 -3.9% vs paid	4.99 -3.1% vs paid	4.99 -13.2% vs paid	4.99 -12.7% vs paid	4.99 -12.8% vs paid	$5.62
Double Cocktail	22.99 +1.4% vs paid	22.99 +0.3% vs paid	30.25 +7.0% vs paid	30.25 +7.2% vs paid	30.99 +9.4% vs paid	$28.79
24oz **Fountain Drink**	6.49 -9.7% vs paid	6.49 -8.1% vs paid	7.79 0.0% vs paid	7.79 +0.8% vs paid	7.79 +0.7% vs paid	$7.79
Bavarian Pretzel	7.49 -7.5% vs paid	7.49 -6.1% vs paid	10.59 0.0% vs paid	10.59 +1.2% vs paid	10.59 +1.5% vs paid	$10.59
16oz **Michelob Ultra**	12.99 +3.3% vs paid	12.99 +2.2% vs paid	17.49 +5.0% vs paid	17.49 +5.4% vs paid	17.99 +8.1% vs paid	$16.99

Last season at these prices

What the 36 past shows would have sold at the prices above. This is not a test: it replays our price assumptions on real past sales, so every group can be compared, including Family Night & Sports. The real test of the groups is the backtest in the notebook.

GROUP	PAST SHOWS	ITEMS SOLD	ITEMS, NEW PRICES	GROSS, REAL	GROSS, NEW PRICES	CHANGE
● Family Day	7	9,697	+5.1%	$72,593	$71,855	-$738 (-1.0%)
● Family Night & Sports	5	14,285	+3.4%	$123,923	$123,331	-$592 (-0.5%)
● Pop	5	19,921	+4.8%	$214,803	$214,970	+$167 (+0.1%)
● Social	10	31,943	+3.0%	$420,409	$423,145	+$2,736 (+0.7%)
● Bar	9	39,460	+2.3%	$518,802	$525,911	+$7,109 (+1.4%)
All 36 shows	**36**	**115,306**	+3.3%	**$1,350,530**	**$1,359,212**	+$8,682 (+0.6%)

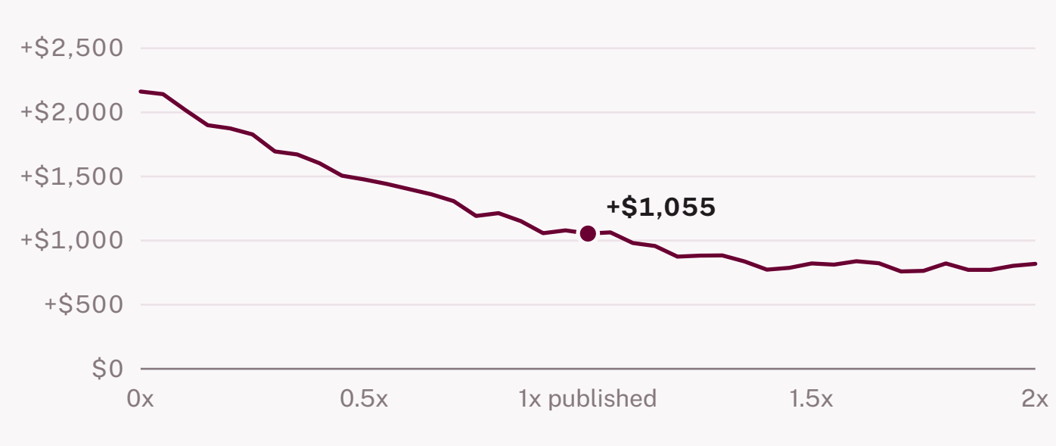
WHAT IF FANS REACT MORE OR LESS TO PRICE?

Base crowd, prices on the card above

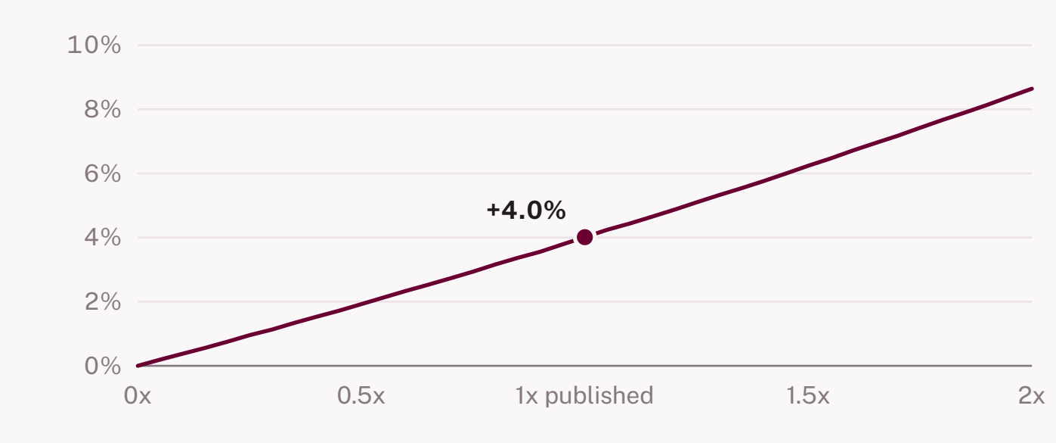
Our data can't measure price sensitivity, because the menu price was the same at every past show. So we borrow it from published studies. These charts run the price card above with sensitivity anywhere from 0x (fans ignore prices) to 2x the published values.

This card makes at least as much as the old prices at every sensitivity from 0x to 2x. If fans ignore prices completely: **+$2,162** (+0.98%). At the published values: **+$1,055** (+0.48%), and items sold go up 4.0%.

Gross sales vs old prices 9 shows, 5 items



Items sold vs old prices 9 shows, 5 items

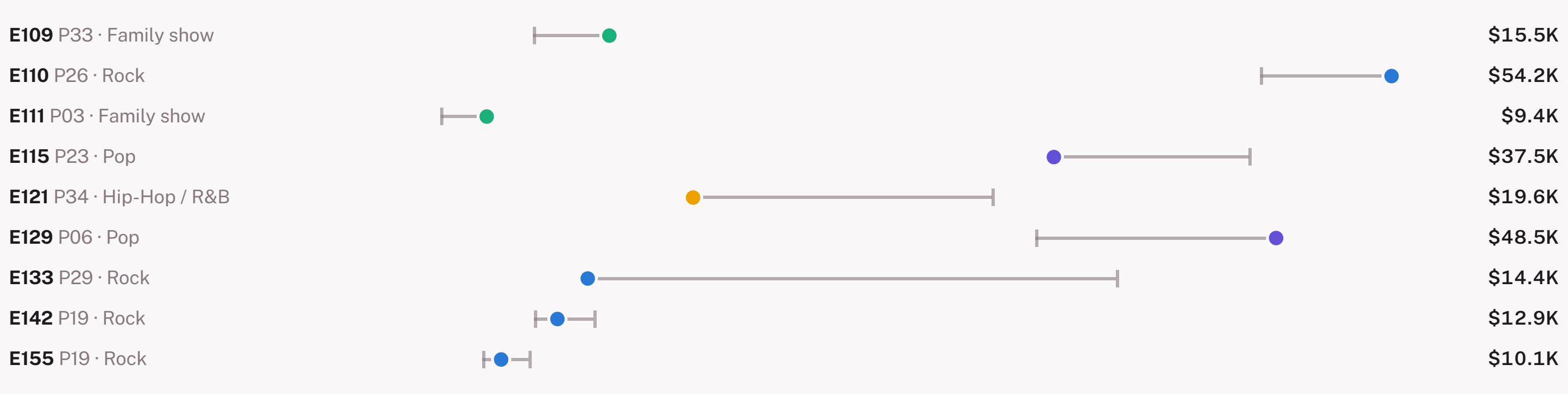


Bottom axis: price sensitivity, where 1x = published studies. The dot is the setting picked at the top. Hover or use the arrow keys on a chart to read any point.

▶ Show the numbers

THE 9 UPCOMING SHOWS

Scale: $0 to $56.4K



Dot: gross for the five items at your prices and the crowd size picked above. Line: low to high crowd.

SHOW	GROUP	CROWD	WATER	COCKTAIL	FOUNTAIN	PRETZEL	MICHELOB	GROSS	VS OLD PRICES	BASED ON
E109 P33, Fri evening	● Family Day	8,830	786	116	491	478	162	$15,460	-$95	Own past shows
E110 P26, Thu evening	● Bar	10,100	1,830	784	400	402	744	$54,186	+$754	Group average
E111 P03, Wed evening	● Family Day	4,300	500	89	393	250	33	$9,393	-$87	Own past shows, evening adjustment from P25
E115 P23, Fri evening	● Pop	10,010	1,876	468	567	465	265	$37,494	+$33	Group average
E121 P34, Sun evening	● Social	5,200	690	341	177	154	163	$19,619	+$142	Group average
E129 P06, Mon evening	● Pop	12,590	2,304	661	665	487	379	$48,459	+$84	Own past shows
E133 P29, Wed evening	● Bar	2,680	486	208	106	107	197	$14,374	+$213	Group average
E142 P19, Tue evening	● Bar	4,490	614	173	170	150	85	$12,867	+$65	Own past shows
E155 P19, Tue afternoon	● Bar	4,540	789	63	151	187	57	$10,072	-$55	Own past shows, matinee adjustment from P25
Total		**62,740**	**9,875**	**2,903**	**3,120**	**2,680**	**2,085**	**$221,923**	+$1,055	

FORECAST FOR EVERY SHOW AND ITEM

Base crowd, sensitivity as published, our card

Projected units, recommended price and projected gross sales for each of the five top items at the 9 upcoming shows. With Our card, Base and As published, these are the numbers in upcoming_forecast.csv.

SHOW	GROUP	ITEM	PROJECTED UNITS	RECOMMENDED PRICE	PROJECTED GROSS
E109 P33	● Family Day	20oz Dasani Water	786	$4.99	$3,922.14
		Double Cocktail	116	$22.99	$2,666.84
		24oz Fountain Drink	491	$6.49	$3,186.59
		Bavarian Pretzel	478	$7.49	$3,580.22
		16oz Michelob Ultra	162	$12.99	$2,104.38
E110 P26	● Bar	20oz Dasani Water	1,830	$4.99	$9,131.70
		Double Cocktail	784	$30.99	$24,296.16
		24oz Fountain Drink	400	$7.79	$3,116.00
		Bavarian Pretzel	402	$10.59	$4,257.18
		16oz Michelob Ultra	744	$17.99	$13,384.56
E111 P03	● Family Day	20oz Dasani Water	500	$4.99	$2,495.00
		Double Cocktail	89	$22.99	$2,046.11
		24oz Fountain Drink	393	$6.49	$2,550.57
		Bavarian Pretzel	250	$7.49	$1,872.50
		16oz Michelob Ultra	33	$12.99	$428.67
E115 P23	● Pop	20oz Dasani Water	1,876	$4.99	$9,361.24
		Double Cocktail	468	$30.25	$14,157.00
		24oz Fountain Drink	567	$7.79	$4,416.93
		Bavarian Pretzel	465	$10.59	$4,924.35
		16oz Michelob Ultra	265	$17.49	$4,634.85
E121 P34	● Social	20oz Dasani Water	690	$4.99	$3,443.10
		Double Cocktail	341	$30.25	$10,315.25
		24oz Fountain Drink	177	$7.79	$1,378.83
		Bavarian Pretzel	154	$10.59	$1,630.86
		16oz Michelob Ultra	163	$17.49	$2,850.87
E129 P06	● Pop	20oz Dasani Water	2,304	$4.99	$11,496.96
		Double Cocktail	661	$30.25	$19,995.25
		24oz Fountain Drink	665	$7.79	$5,180.35
		Bavarian Pretzel	487	$10.59	$5,157.33
		16oz Michelob Ultra	379	$17.49	$6,628.71
E133 P29	● Bar	20oz Dasani Water	486	$4.99	$2,425.14
		Double Cocktail	208	$30.99	$6,445.92
		24oz Fountain Drink	106	$7.79	$825.74
		Bavarian Pretzel	107	$10.59	$1,133.13
		16oz Michelob Ultra	197	$17.99	$3,544.03
E142 P19	● Bar	20oz Dasani Water	614	$4.99	$3,063.86
		Double Cocktail	173	$30.99	$5,361.27
		24oz Fountain Drink	170	$7.79	$1,324.30
		Bavarian Pretzel	150	$10.59	$1,588.50
		16oz Michelob Ultra	85	$17.99	$1,529.15
E155 P19	● Bar	20oz Dasani Water	789	$4.99	$3,937.11
		Double Cocktail	63	$30.99	$1,952.37
		24oz Fountain Drink	151	$7.79	$1,176.29
		Bavarian Pretzel	187	$10.59	$1,980.33
		16oz Michelob Ultra	57	$17.99	$1,025.43
All 9 shows			**20,663**		**$221,923.07**

EVERY SHOW AND ITS GROUP

36 past shows and the 9 upcoming shows (in bold)

● Family Day 7 past, 2 upcoming
Past: E107, E114, E118, E124, E130, E138, E150
Upcoming: **E109**, **E111**

● Family Night & Sports 5 past, 0 upcoming
Past: E101, E108, E125, E131, E151
Upcoming: none on the schedule yet

● Pop 5 past, 2 upcoming
Past: E116, E120, E127, E136, E145
Upcoming: **E115**, **E129**

● Social 10 past, 1 upcoming
Past: E103, E105, E113, E117, E134, E141, E144, E147, E148, E154
Upcoming: **E121**

● Bar 9 past, 4 upcoming
Past: E104, E106, E112, E122, E126, E132, E135, E140, E143
Upcoming: **E110**, **E133**, **E142**, **E155**

PLAN A NEW SHOW

Our grouping rule on a show that isn't on the schedule yet

Event type
Concert

Genre
Country

Start time
Evening

Expected crowd
11000

● Bar A rock or country concert goes to Bar. Past shows in this group spent $13.3 a head on alcohol.

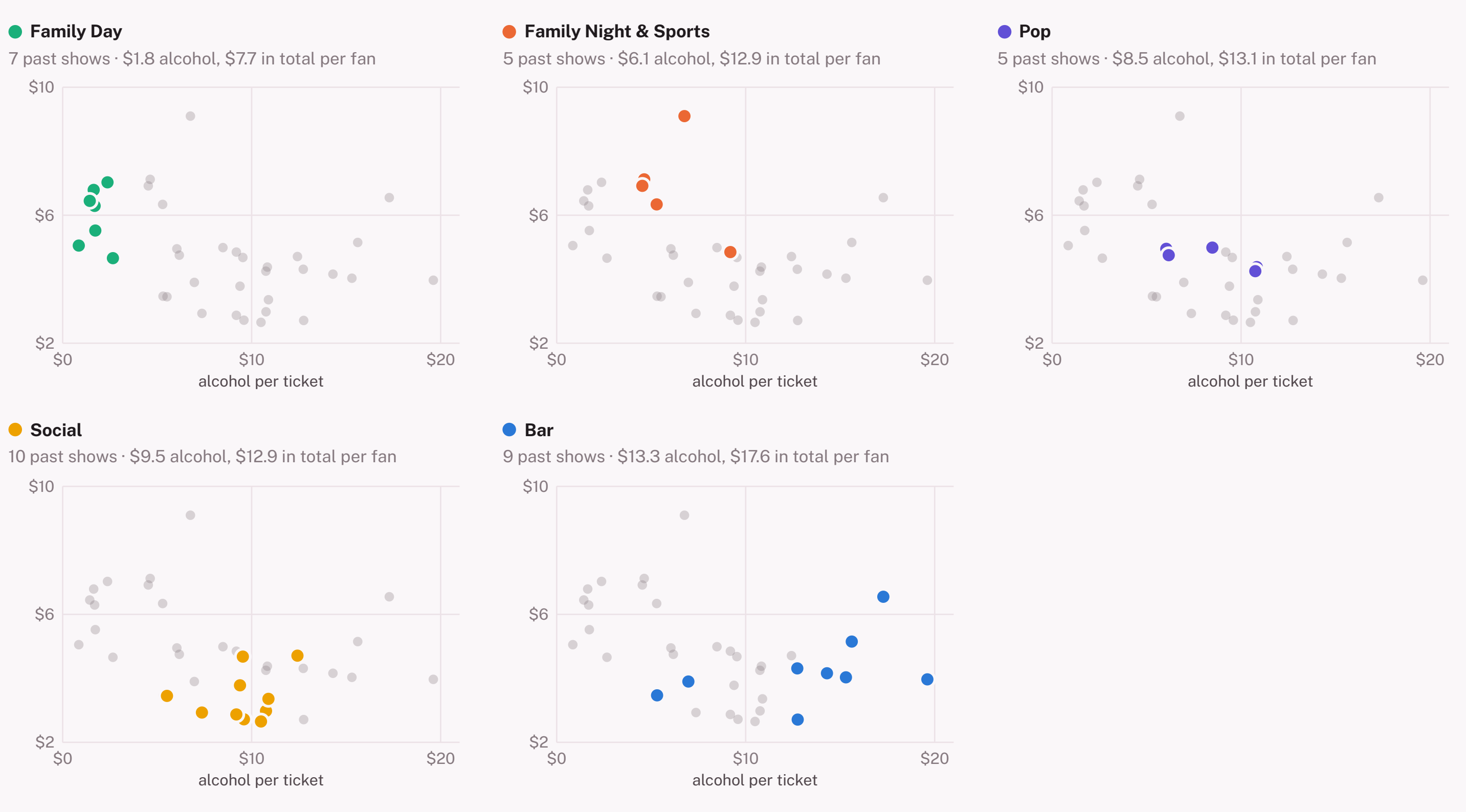
ITEM	PRICE	UNITS	GROSS
Dasani Water	$4.99	1,993	$9,945
Double Cocktail	$30.99	854	$26,465
Fountain Drink	$7.79	435	$3,389
Bavarian Pretzel	$10.59	438	$4,638
Michelob Ultra	$17.99	810	$14,572
Total		**4,530**	**$59,010**

That is $5.36 per fan on these five items. Prices come from the price card above.

PAST SHOWS BY GROUP

36 shows, dollars per ticket sold

Each panel lights up one group; the gray dots are all the other past shows, for comparison. Across: alcohol per ticket. Up: food and soft drinks per ticket. Hover a colored dot for the show.


● Family Day 7 past shows · $1.8 alcohol, $7.7 in total per fan

● Family Night & Sports 5 past shows · $6.1 alcohol, $12.9 in total per fan

● Pop 5 past shows · $8.5 alcohol, $13.1 in total per fan

● Social 10 past shows · $9.5 alcohol, $12.9 in total per fan

● Bar 9 past shows · $13.3 alcohol, $17.6 in total per fan

▶ Show the numbers

HOW THE NUMBERS ARE BUILT

- **Units** = crowd x units per ticket x (new price / price paid before)$^{\text{elasticity}}$. **Gross** = units x price.
- **Last season at these prices** takes each past show's real units and sales and re-prices them with the same formula, using what that show actually paid as the old price. It shows what our assumptions imply. It does not check them.
- **The real test** is the backtest in the notebook: hide one act's shows, predict them from the other shows in the same group, and compare with what they actually sold. Error: 34% with no groups, 24% with our first 4 groups, 22% with the 5 groups (pop on its own).
- **Price sensitivity check:** menu prices never changed in the data, so we can't measure sensitivity ourselves. We tried estimating it from the small differences in what fans paid within each group, and it was too noisy to use (95% interval from -1.84 to +3.97). The charts above re-run the forecast from 0x to 2x the published values instead.
- **Units per ticket** come from the performer's own past shows when they played here before (P03, P06, P19, P33), otherwise from their group. E111 and E155 change between daytime and evening, so they get P25's daytime/evening ratios.
- **Crowd:** base = Rule 2. High = the largest of Rules 1 to 3. Low = the smallest, except repeat acts get 75% of their past crowd, never below tickets already sold.
- **Elasticities** are not in our data, so we use published averages: beer -0.46, spirits -0.80 (Wagenaar et al. 2009), soft drinks -0.79 and food away from home -0.81 (Andreyeva et al. 2010). The sensitivity switch scales all of them.
- **Groups:** sports event or a family show where at least half the buyers are men = Family Night & Sports; other family shows = Family Day; rock or country concert = Bar; pop concert = Pop; everything else (hip-hop/R&B, comedy) = Social. Pop and Social keep the same prices; they are separate because pop crowds buy a different mix.
- **Tools:** Python (pandas, NumPy, SciPy, scikit-learn, matplotlib) for the analysis, HTML and JavaScript for this dashboard, and Streamlit with Altair for the live app.
- Concessions data covers most but not all points of sale, so treat totals as a consistent measure, not the arena's full revenue.